

Project Design Phase-II Technology Stack (Architecture & Stack)

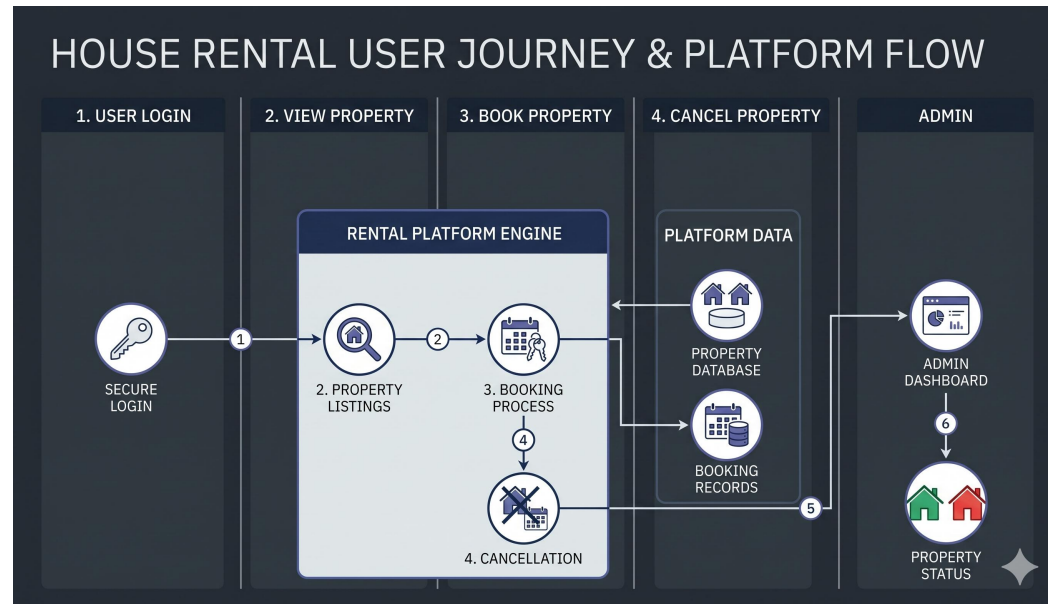
Date	18 June 2026
Project Name	HouseRentals
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Booking A property

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI.	HTML, CSS, JavaScript / / React Js etc.
2.	Application Logic-1	Logic for a process in the application	Nodejs
3.	Database	Data Type, Configurations etc.	NoSQL(MongoDB)
4.	Cloud Database	Database Service on Cloud	MongoDB
5.	File Storage	File storage requirements	Cloudinary
6.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration :	Local, Cloud Foundry, github, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used	Technology of Opensource framework
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	Technology used
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	Technology used
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Technology used

References:

<https://www.freecodecamp.org/learn/javascript-algorithms-and-data-structures/>

<https://react.dev/>

<https://nodejs.org/en/docs/>

<https://expressjs.com/>

<https://learn.mongodb.com/>

<https://www.npmjs.com/package/jsonwebtoken>

<https://cloudinary.com/documentation>